

class notes

2026-05-16

Fractions and Ratios

Learning objectives

- Identify equivalent fractions.
- Explain the relationship between ratios and fractions.
- Apply the rules for comparing and adding fractions.
- Contrast the concepts of ratios and fractions.
- Predict the result of multiplying fractions.

Key concepts

Equivalent Fractions

- Equivalent fractions are fractions that represent the same amount.
- They can be found by multiplying or dividing both the numerator and denominator by the same non-zero number.
- The fraction $\frac{1}{2}$ is equivalent to $\frac{2}{4}$ and $\frac{4}{8}$.

Ratios

- A ratio compares two quantities.
- It can be written as A:B or as $\frac{A}{B}$.
- The ratio 6:2 simplifies to 3:1.

Fractions

- A fraction is a part of a whole, represented by the numerator and denominator.
- The numerator indicates the number of parts you have, and the denominator indicates the total number of equal parts in the whole.
- The fraction $\frac{3}{4}$ means three out of four equal parts.

Worked example

- The ratio of apples to oranges in a basket is 6:2, which simplifies to 3:1. If the basket contains 6 apples and 2 oranges, what is the ratio of apples to oranges? The ratio is 6:2, which simplifies to 3:1. This demonstrates the concept of simplifying ratios.

Common misconceptions

- Adding fractions means adding the numerators and the denominators separately.
- A bigger denominator always means a bigger fraction.
- Ratios, fractions, and percentages are three views of the same idea — the relationship between a part and a whole.